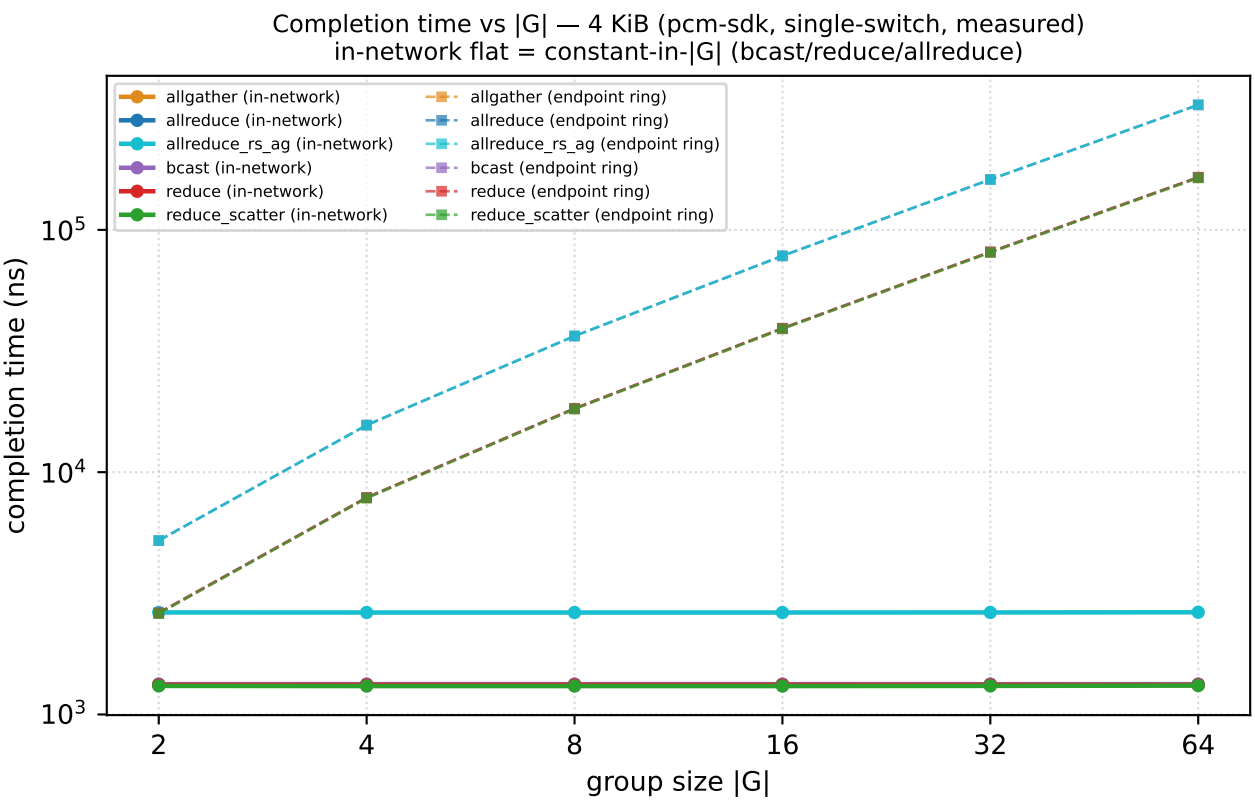
Completion time vs |G| — 4 KiB (pcm-sdk, single-switch, measured)
in-network flat = constant-in-|G| (bcast/reduce/allreduce)
allgather (in-network)
allreduce (in-network)
allreduce_rs_ag (in-network)
bcast (in-network)
reduce (in-network)
reduce_scatter (in-network)
allgather (endpoint ring)
allreduce (endpoint ring)
allreduce_rs_ag (endpoint ring)
bcast (endpoint ring)
reduce (endpoint ring)
reduce_scatter (endpoint ring)
completion time (ns)
10^5
10^4
10^3
2
4
8
16
32
64
group size |G|